

Methodology and Figures Overview

Methodology

Our project focuses on understanding the VOC workforce between 1700 and 1780. To do this, we first cleaned and organized the contracts dataset so that we could use it for analysis and visualizations later on.

Data Cleaning

The first step was to filter the dataset to only include contracts that started between 1700 and 1780.

We then made sure that each contract contained useful information such as the person's region of origin, their rank in the company, and the outcome of their contract (for example, whether they died, were repatriated, or left the company).

To make the dataset easier to work with, we grouped places of origin into broader regions, like the Dutch Republic, Low German area, British Isles, and so on. If the region was missing but the place name looked Dutch (for example, Amsterdam or Zeeland), we assigned it to the Dutch Republic.

Ranks were grouped into six larger categories: Sea, Ship, Trade, Medical, Military, and Other. We also estimated a level of seniority based on wages, so that we could later compare higher and lower positions.

The outcomes were simplified into four clear groups: Death, Repatriated, Attrition (which means people who left, deserted, or were dismissed), and Unknown.

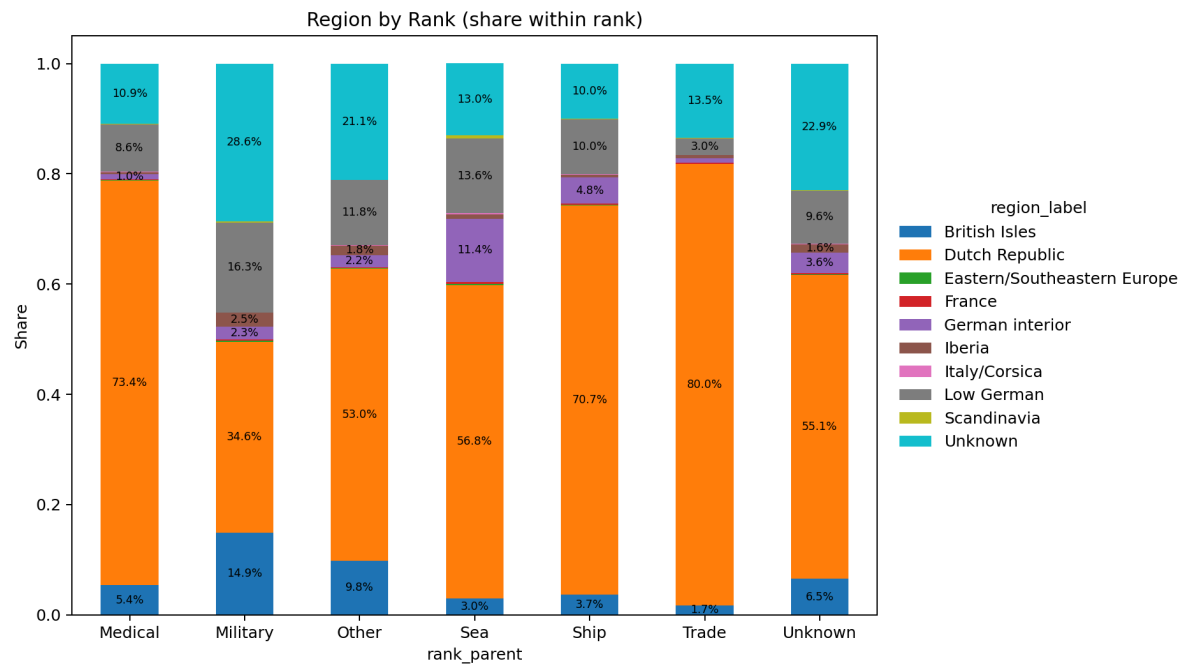
Finally, we marked which contracts were a person's first one, so that we could identify recruitment patterns.

After this cleaning process, the dataset was saved in a clean format and summary tables were created for regions, ranks, and outcomes.

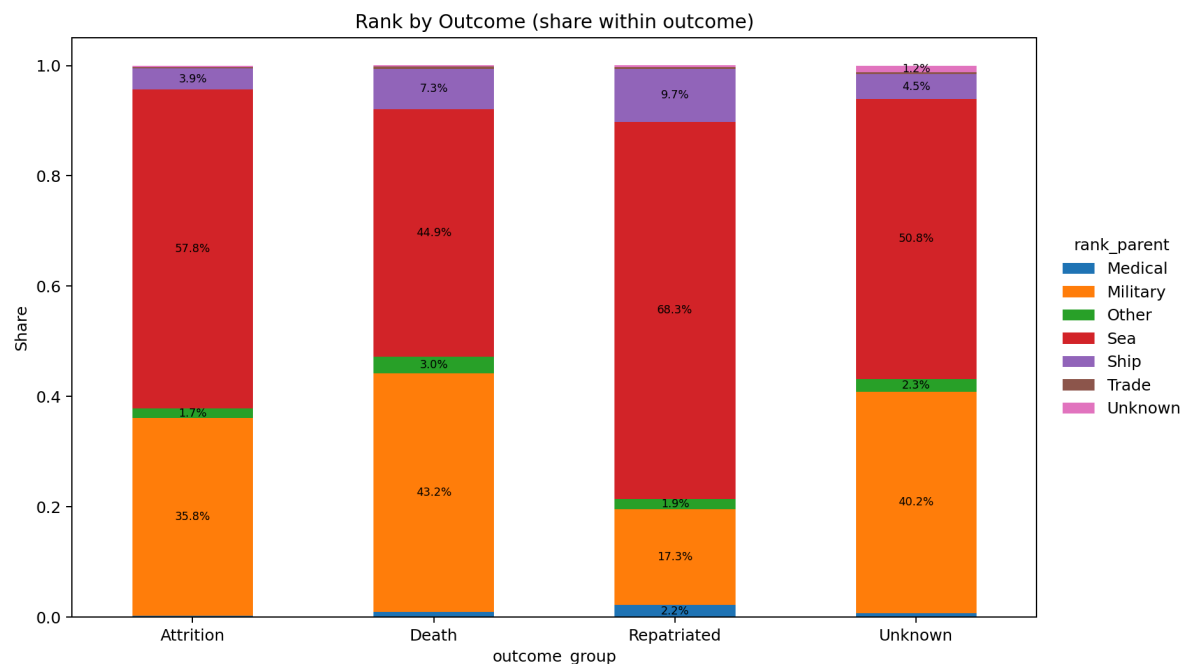
Descriptive Analysis and Visualizations

Once the data was ready, we created several visualizations to explore trends and relationships.

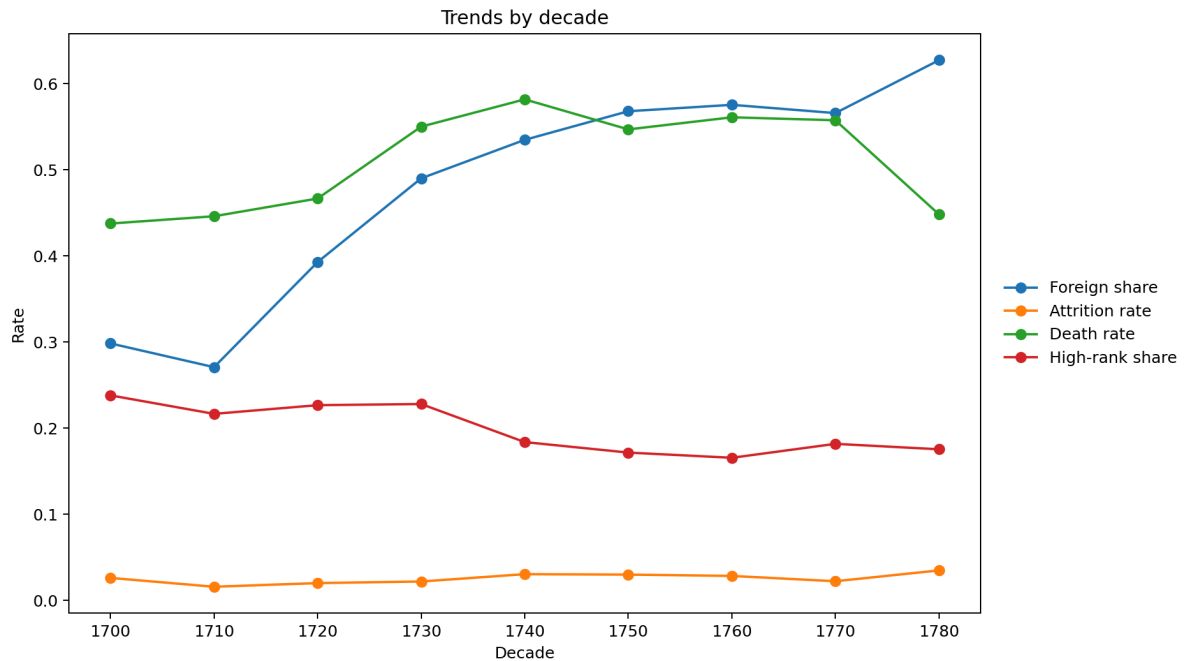
Region by Rank: shows how people from different regions were distributed across the various ranks. This helps us see whether Dutch workers were more concentrated in higher positions or if certain regions were mostly represented in lower ranks.



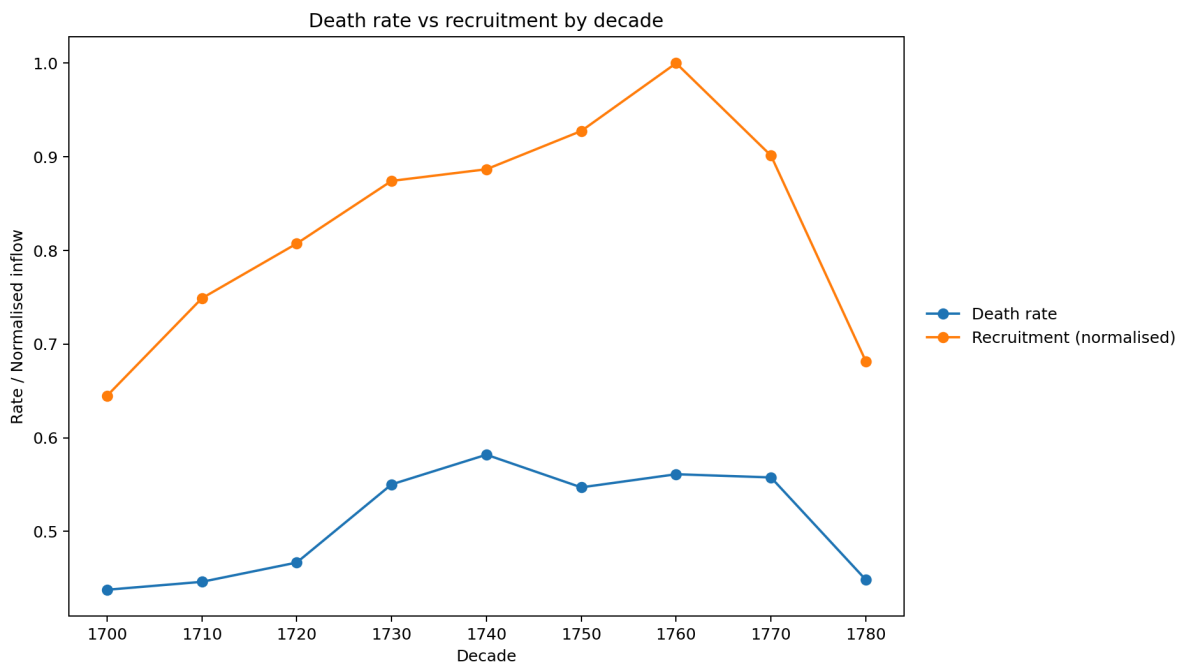
Rank by Outcome: shows how contract outcomes differ between ranks. It helps us see which roles had higher death rates or which were more likely to be repatriated.



Trends by Decade: tracks how the share of foreign workers, the death and attrition rates, and the share of high-ranking positions changed over time.



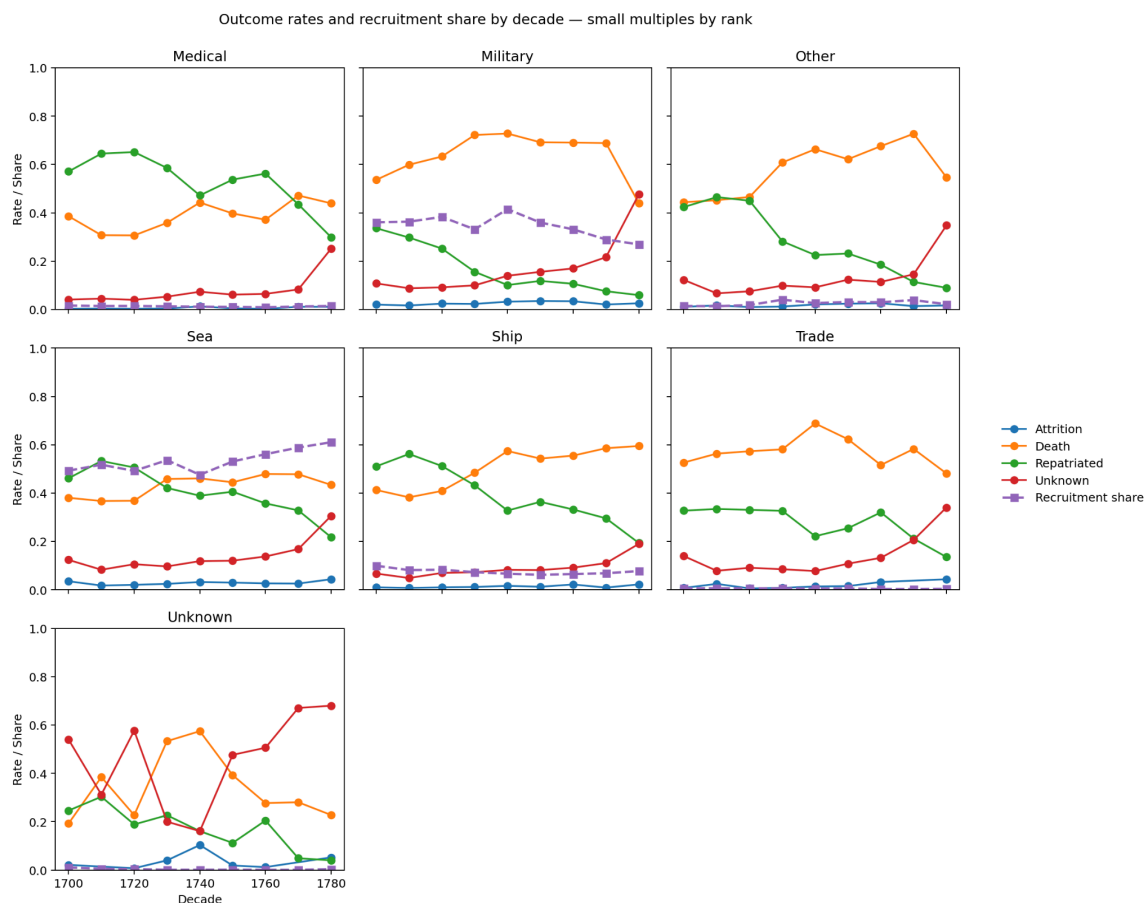
Death Rate vs. Recruitment: compares how many people died with how many new workers were being recruited each decade. This helps us see whether the VOC was able to replace losses.



Deaths vs. Recruitment by Rank: compares, for each rank, the rate of death and the share of recruitment. For example, if a rank shows high death but low recruitment, it could suggest that replacements were not keeping up.



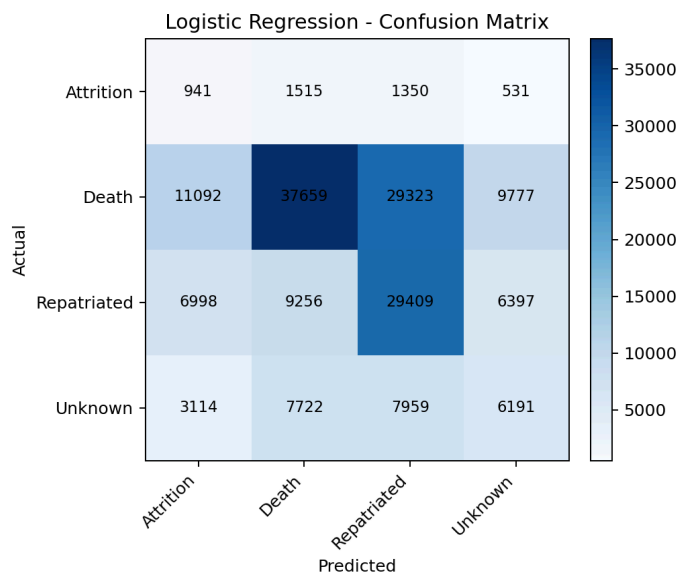
Outcome Trends and Recruitment by Rank: this figure shows each rank in its own small panel. Each panel tracks how outcomes and recruitment changed over time. It helps identify which ranks faced the most risk and where the company focused its recruitment efforts.



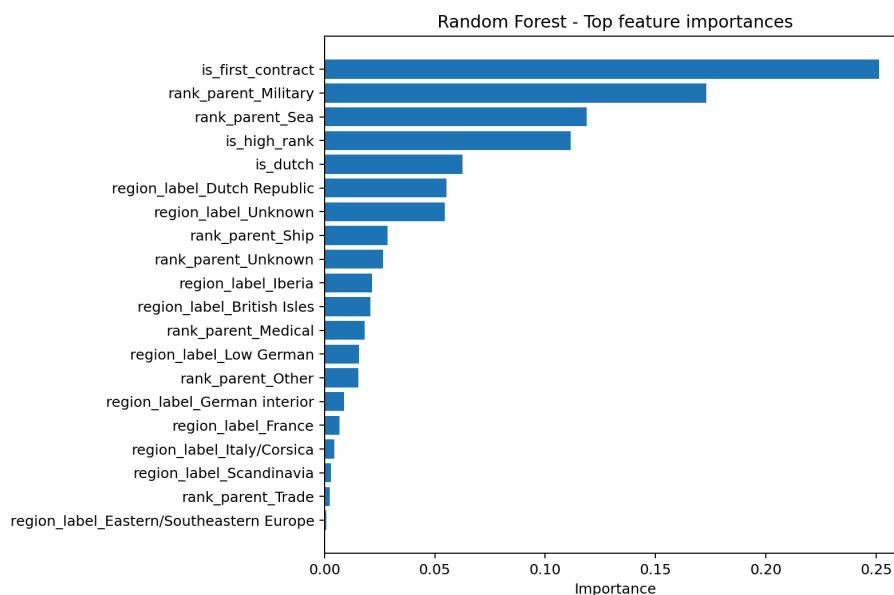
Modeling

Finally, we built two simple models to see whether we could predict contract outcomes based on basic characteristics like region, rank, decade, and whether someone was Dutch, high-ranking, or newly recruited.

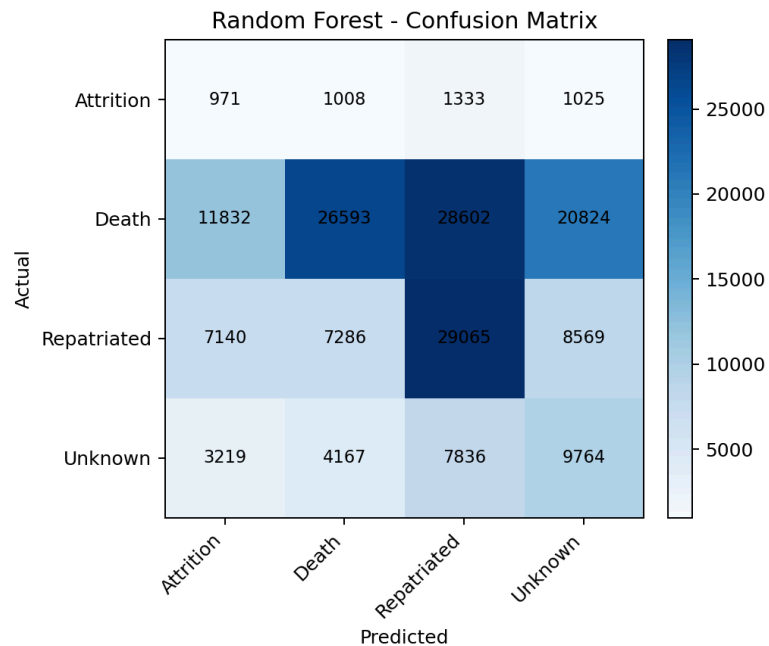
The **Logistic Regression** model gives a general picture of how different factors might influence the likelihood of each outcome. It's a simpler, more interpretable model that treats each outcome as a separate comparison. The confusion matrix for this model shows how often the predictions matched the actual outcomes.



The Random Forest model looks for more complex patterns. It tends to perform better on uneven data, because it builds many small decision trees and averages their results. The **feature importance** chart shows which factors mattered most when predicting outcomes, for example, rank, region, or being Dutch.



The confusion matrix for the **Random Forest** model also shows how well it predicted each category of outcome.



Summary

In short, the cleaning step prepared the data and grouped it into comparable categories. The descriptive analysis helped us see patterns between origin, rank, outcome, and time. The modeling part provided an idea of which characteristics were most strongly linked to the different outcomes.

Together, these steps form the full methodological process that connects raw historical data to meaningful insights about how the VOC workforce functioned over the eighteenth century.